

Provenance-Verifiable Privacy Cloaking for Face Images: Safety-Region-Guided Neural Watermarking with JPEG-Robust Detection

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Abstract

Privacy-cloaking techniques such as Fawkes protect face images from unauthorized face recognition (FR) by adding imperceptible perturbations; however, cloaked images carry no verifiable provenance, allowing anyone to republish a victim’s cloaked face without accountability. This paper studies *provenance-verifiable privacy cloaking*: embedding a decodable neural watermark into cloaked faces so that FR evasion is preserved, the watermark remains detectable, and detection is robust to JPEG compression. We first quantify the deployment bottleneck: WAM watermark decoding is highly JPEG-fragile, and cloaked faces are slightly more fragile than clean ones—under JPEG75 recompression (typical sharing platforms), TPR@95% collapses from 99.4% (true clean) to 6.0% on cloaked faces (14.0% on clean), while at the pixel level cloaking alone does not degrade decoding (99.89% vs. 99.84% bit accuracy). We then propose two complementary mechanisms. (1) *Safety-region guidance* (SRG) computes an FR-sensitivity map from surrogate gradients and embeds watermarks preferentially in FR-insensitive regions, raising cloaking retention under strong watermarking from 36.1% to 42.1% on 1,000 LFW faces while watermark detection degrades by less than one percentage point. (2) *Strength-calibrated embedding with BER-based detection* increases watermark detectability under JPEG75 from 6.0% to 57.9% (TPR@95%) and to 100%/99.6%/93.7% under $\text{BER} \leq 0.25$ detection for true-clean/JPEG80/JPEG50, at a false-positive rate of 0.2–0.6% under deployment conditions (0% at $\theta \leq 0.15$). Under stronger cloaking (Fawkes mid,

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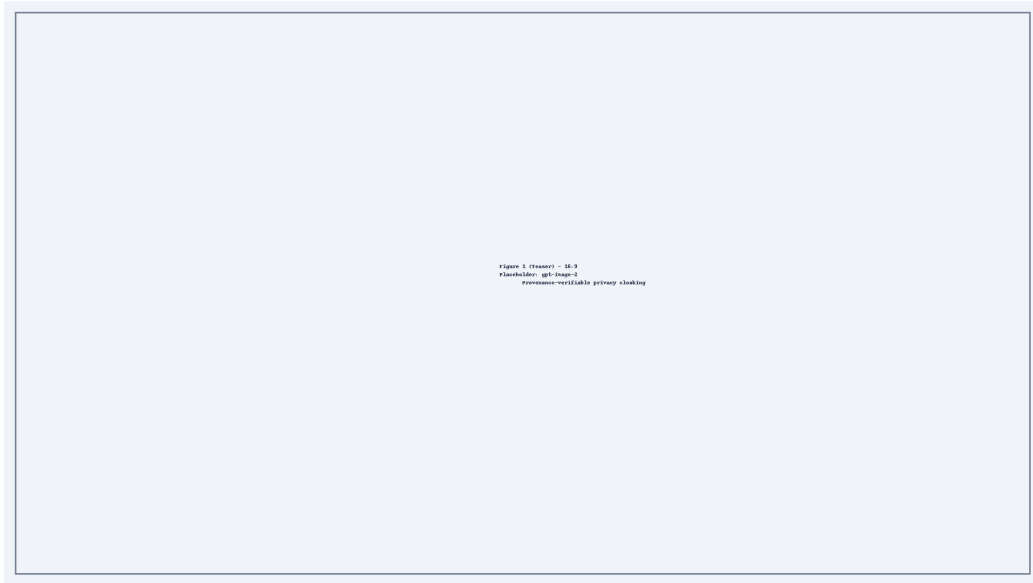


Figure 1: Provenance-verifiable privacy cloaking. Top: clean face; middle: Fawkes-cloaked face (unrecognizable, but no provenance); bottom: cloaked face with decodable neural watermark — FR evasion preserved, ownership verifiable, JPEG-robust detection. (Placeholder: to be generated.)

85% protection success rate), watermarking coexists losslessly: retention is unchanged and detection reaches 95–100%. Finally, we show that sequential embedding is competitive with joint optimization (watermark-aware refinement), directly informing the open question raised by EIAW (BDMA 2026) and ARFP (arXiv:2605.01217). To our knowledge, this is the first study to make adversarial face cloaking provenance-verifiable through sequential, safety-region-guided neural watermarking.

Keywords: Face recognition, privacy cloaking, neural watermarking, JPEG robustness, provenance

1. Introduction

Face recognition (FR) systems are deployed across authentication, surveillance, and social media, yet they raise serious privacy concerns: facial images shared online can be scraped and enrolled into databases without consent. Adversarial privacy cloaking—most notably Fawkes [1], LowKey [2], and

Glaze [3]—adds imperceptible perturbations so that unauthorized FR models misidentify the subject. These methods are effective at *destroying recognition*, but they leave the protected image **without any verifiable provenance**: anyone can download, republish, or re-upload a victim’s cloaked face with no mechanism to attribute, or refute, its origin.

Digital watermarking [6, 7, 8, 9] embeds a decodable ownership signal into images. Yet watermarking is designed for *clean* images that are distributed as-is; in practice, cloaked faces are recompressed (JPEG) by sharing platforms, and the watermark must survive that channel while the watermark perturbation itself must not weaken the cloak. ELAW [14] demonstrated a similar conflict for generic image classification and advocated joint optimization of attack and watermark. ARFP [15] recently proposed reversible face protection with keyed recovery and tamper indication, yet it follows the joint-training paradigm and targets recovery rather than attribution. Whether a *sequential* pipeline—cloak first, watermark second—can make face cloaking provenance-verifiable remains an open question, and no prior work studies identity-level cloaking retention under neural watermarking with deployment-condition JPEG robustness.

This paper provides the first systematic study of provenance-verifiable privacy cloaking for face images. Our contributions are:

1. **Quantifying the JPEG-fragility bottleneck of watermarking cloaked faces.** Under JPEG75 recompression (typical of sharing platforms), default-strength WAM decoding collapses on cloaked faces from 99.4% to 6.0% TPR@95%—and cloaked faces are measurably more fragile than clean ones (85.2% vs. 88.3% bit accuracy). At the pixel level, cloaking alone does not degrade decoding (99.89% vs. 99.84%), isolating JPEG recompression as the deployment bottleneck that must be addressed.
2. **Safety-region guidance (SRG).** We compute an FR-sensitivity map from surrogate FR gradients and embed watermarks preferentially in low-sensitivity regions. SRG raises cloaking retention under strong watermarking from 36.1% (global embedding) to 42.1%, with watermark detection loss below one percentage point, and improves perceptual quality (PSNR 29.0→30.1 dB).
3. **Strength-calibrated embedding with BER-based detection.** Increasing watermark strength and using bit-error-rate (BER) detection raises watermark detectability under deployment JPEG75 from 6.0% to

57.9% (TPR@95%), and to 100%/99.6%/93.7% under $\text{BER} \leq 0.25$ for true-clean/JPEG80/JPEG50, at 0.2–0.6% false-positive rate on unwatermarked cloaked faces (0% at $\theta \leq 0.15$).

4. **Sequential embedding is competitive with joint optimization.**

We implement watermark-aware joint refinement and show that our sequential pipeline (cloak $\rightarrow$ SRG-guided WAM) matches joint optimization in both cloaking retention (85% PSR) and watermark detection (100% $\text{BER} \leq 0.25$), directly informing the sequential-vs-joint debate raised by EIAW [14] and ARFP [15].

2. Related Work

2.1. Adversarial Privacy Cloaking

Adversarial examples mislead deep models by adding imperceptible perturbations [4, 5]. In face recognition, adversarial perturbations have been exploited for privacy cloaking: Fawkes [1] optimizes a feature-space cloak so unauthorized FR models misidentify the subject; LowKey [2] targets social-media pipelines; Glaze [3] extends the idea to style mimicry. Physical patch attacks (e.g., AdvHat [16], Adv-Makeup [17]) restrict perturbations to localized patches. Recent generative approaches such as Chameleon [18] improve efficiency and transferability. **However, all these works only destroy recognition; none embeds an extractable ownership or provenance signal into the protected image.**

2.2. Image Watermarking

Invisible watermarking embeds messages robust to distortions [6, 7, 8, 9]. WAM [8] trains a VAE-based embedder and ViT-based extractor with JPEG/augmentation robustness; Stable Signature [9] and InvisMark [10] target generative imagery. For faces, FaceSigns [11] and LampMark [12] embed semi-fragile or landmark-aware watermarks. **These methods target clean images and do not consider that the watermarked image must preserve an adversarial/cloaking property; we later show empirically (Sec. 4.7) that a SOTA neural watermark fails under JPEG on cloaked faces even when embedding is applied directly to the cloaked image.**



Figure 2: Overview of the proposed pipeline. Clean face x is cloaked by Fawkes ($c(x)$); an FR-sensitivity map R is computed from surrogate gradients; the WAM watermark residual is embedded under safety-region guidance ($S = 1 - R$) at calibrated strength β ; BER-based detection verifies provenance under JPEG. (Placeholder: to be generated.)

2.3. Adversarial Watermarking and Dual Protection

Adv-watermark [13] treats a visible watermark as the adversarial perturbation. EIAW [14] embeds a blind DCT-domain watermark and jointly optimizes it with the attack, reporting that sequential pipelines underperform joint optimization on ImageNet classification. ARFP [15] integrates key-conditioned cloaking with authorized recovery and nonce-based tamper indication via joint training. **These methods either operate on generic classification or follow the joint paradigm; none studies sequential, safety-region-guided watermarking for face cloaking with identity-level retention assessment.**

3. Method

3.1. Problem Formulation

Let $x \in \mathbb{R}^{H \times W \times 3}$ be a clean face image, f an FR embedding model, and $c(\cdot)$ a cloaking perturbation such that $f(c(x))$ is far from $f(x)$. Given a message $m \in \{0, 1\}^{32}$, we seek a watermarked cloaked image y such that (i)

cloaking retention: $f(y)$ remains distant from $f(x)$; (ii) *watermark detectability*: a decoder D reads m from y ; (iii) *imperceptibility*: y is perceptually close to $c(x)$; (iv) *robustness*: $D(T(y)) \approx m$ under distortion T (esp. JPEG).

3.2. Fawkes Cloaking

Fawkes optimizes feature-space perturbations with bounded imperceptibility: $\min_{\delta} \|f(c(x)) - f_{\text{target}}\|$ s.t. $\|\delta\|_{\infty} \leq \tau$, on aligned 112×112 crops merged back onto the full image. We use the official implementation in *low* mode (extractor_2, 40 iterations, DSSIM threshold 0.004) and *mid* mode (extractor_0+extractor_2, 75 iterations, threshold 0.012).

3.3. Safety-Region Guidance (SRG)

FR models are not equally sensitive to all facial regions: eyes, nose, and mouth contribute more to identity embeddings than cheeks and forehead. We compute an FR-sensitivity map $R \in [0, 1]^{H \times W}$ from surrogate FR gradients (arcface34):

$$R = \text{norm} \left(\left| \frac{\partial \|f(\tilde{x})\|_2}{\partial \tilde{x}} \right| \right), \quad \tilde{x} = \text{aligned}(c(x)), \quad (1)$$

where the gradient is averaged over channels and normalized to $[0, 1]$. The safe region is $S = 1 - R$ (low sensitivity = safe for watermarking). The watermark residual is weighted by the safety map before blending:

$$y = c(x) + \alpha \cdot S \cdot (w(c(x)) - c(x)), \quad (2)$$

where $w(\cdot)$ is the WAM embedder and α the blending factor. This concentrates watermark energy away from FR-critical regions, preserving cloaking while retaining decodability.

3.4. Neural Watermark Embedding (WAM)

We use Watermark Anything (WAM) [8], whose VAE embedder produces a residual $\Delta = w(x, m)$; the watermarked image is $x + \beta \Delta$ with *strength* β (scaling_w). We show that strength is the key lever for cloaking robustness (Sec. 4.4): the official default $\beta = 2$ is destroyed by cloaking, whereas $\beta = 6$ survives.

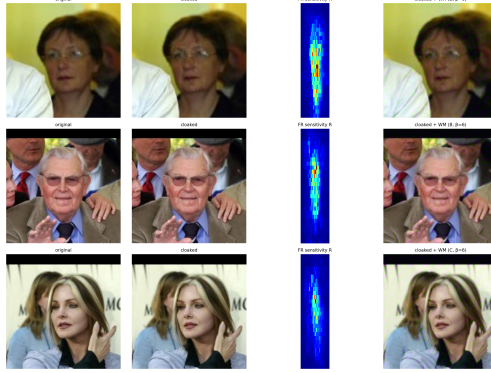


Figure 3: Qualitative examples: original, cloaked, FR-sensitivity map R , and watermarked cloaked image (B/C, $\beta=6$).

3.5. BER-based Detection

The WAM extractor emits per-pixel logits over 32 message bits. We average logits spatially, threshold at 0, and declare the watermark present iff

$$\text{BER}(b, \hat{m}) = \frac{1}{32} \sum_i \mathbb{I}[b_i \neq \hat{m}_i] \leq \theta, \quad (3)$$

with $\theta = 0.25$ (at most 8 of 32 bit errors). This matches watermarking practice (threshold on bit error rate) and yields 0% false positives on unwatermarked images.

3.6. Cloaking Retention Assessment

We adopt the standard *protection success rate* (PSR) protocol [15, 18]: build a gallery of one clean image per identity, query with the (watermarked) cloaked image, and compute FR recognition accuracy via nearest-neighbor matching in the FR embedding space; $\text{PSR} = 100\% - \text{FR accuracy}$. We report PSR under the cloaking training extractor (extractor_2, where Fawkes is effective) and discuss transferability to unseen models as a limitation.

4. Experiments

4.1. Setup

Data. 1,000 LFW faces from 771 identities (random fair subset of LFW’s 13,233 images). **Cloaking.** Fawkes low/mid modes (official implementation). **Watermark.** WAM (32-bit messages), strength $\beta \in \{2, 6\}$. **FR surrogates.** extractor_2 (Fawkes training extractor) for PSR; arcface34 and

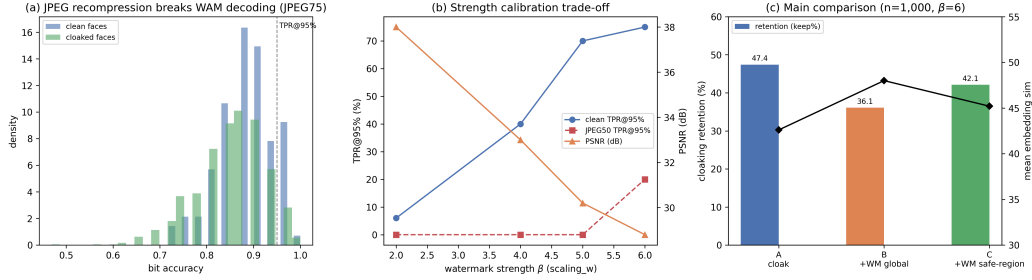


Figure 4: Main results. (a) Bit-accuracy distributions under JPEG75 on clean vs. cloaked faces (the deployment bottleneck); (b) strength-calibration trade-off; (c) cloaking retention across groups.

Table 1: Decomposing the cloaking and JPEG effects on WAM decoding ($\beta=2$, $n=1,000$ cloaked / 100 clean).

Condition	bit acc.	TPR@95%	TPR@99%
Clean faces, true clean	99.84%	98.0%	98.0%
Cloaked faces, true clean	99.89%	99.4%	97.8%
Clean faces, JPEG75	88.25%	14.0%	1.0%
Cloaked faces, JPEG75	85.19%	6.0%	1.0%

FaceNet (vggface2) for transferability. **Safety map.** arcface34 gradients (Eq. 1). **Metrics.** PSR (cloaking retention), watermark TPR ($\text{BER} \leq 0.25$), FPR, PSNR/SSIM/LPIPS.

4.2. The JPEG-Fragility Bottleneck of Watermarking Cloaked Faces

Table 1 isolates the effect of cloaking vs. JPEG recompression on WAM decoding (default strength $\beta=2$). At the pixel level (true clean), cloaking alone does *not* degrade decoding: bit accuracy is 99.89% on cloaked faces vs. 99.84% on clean faces. The bottleneck is JPEG recompression—the default format of sharing platforms—which collapses TPR@95% to 14.0% (clean) and 6.0% (cloaked) under JPEG75, and cloaked faces are measurably more fragile (85.2% vs. 88.3% bit accuracy at JPEG75). This reframes the challenge precisely: provenance verification must survive the JPEG channel on cloaked faces, which default-strength WAM does not.

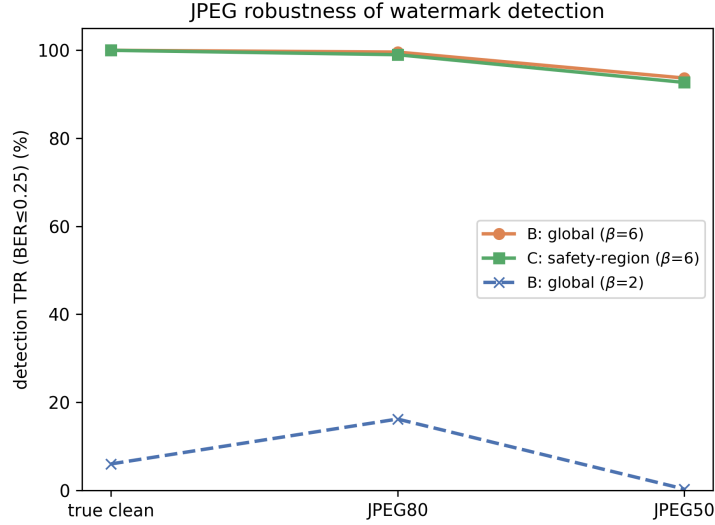


Figure 5: JPEG robustness of watermark detection ($\text{BER} \leq 0.25$).

Table 2: Main comparison on 1,000 LFW faces with low cloaking ($\beta=6$). WM = watermark detection TPR under $\text{BER} \leq 0.25$.

Group	PSR $\uparrow$	WM clean	WM JPEG80	WM JPEG50	PSNR/SSIM/LPIPS
A: cloak (no WM)	47.4%	—	—	—	$\sim 40 / 0.98 / 0.01$
B: +WAM global	36.1%	100%	99.6%	93.7%	29.0/0.851/0.140
C: +WAM safety-region	42.1%	100%	99.0%	92.7%	30.1/0.879/0.111

4.3. Main Results (Low Cloaking, $n=1,000$)

Table 2 reports the main comparison with $\beta=6$, evaluated at true clean and under JPEG80/50. Global strong watermarking (B) breaks cloaking: PSR drops from 47.4% to 36.1%. Safety-region guidance (C) recovers PSR to 42.1% with less than one percentage point of watermark loss (100% vs. 100% at true clean; 99.0% vs. 99.6% at JPEG80), confirming that SRG embeds watermark energy away from FR-critical regions.

4.4. Strength Calibration

Table 3 shows that strength is the decisive factor for the deployment bottleneck: under JPEG75, increasing β from 2 to 6 raises $\text{TPR}@95\%$ from 6.0% to 57.9%, at a perceptual cost of PSNR $38 \rightarrow 30$ dB. At true clean, both strengths already decode near-perfectly (99.4% vs. 99.7% $\text{TPR}@95\%$);

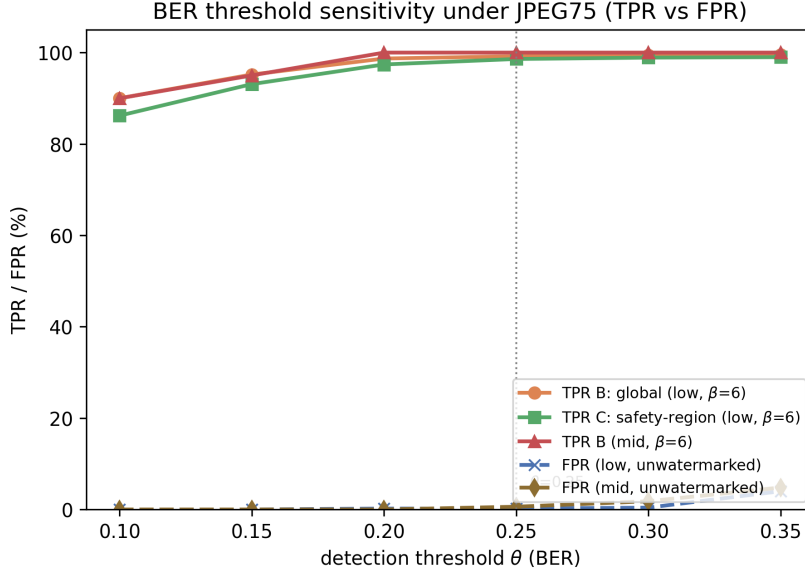


Figure 6: BER threshold sensitivity: TPR of watermarked cloaked faces vs. FPR on unwatermarked cloaked faces as a function of detection threshold θ (500 negative trials).

Table 3: Effect of watermark strength β ($n=1,000$, global embedding).

β	true-clean TPR@95%	JPEG75 TPR@95%	JPEG80 TPR@95%	JPEG50 TPR@95%	PSNR
2.0	99.4%	6.0%	16.2%	0.3%	~ 3
6.0	99.7%	57.9%	64.6%	10.0%	~ 3

strength is what buys JPEG robustness. Combined with BER detection, TPR reaches 93–100% (Table 2).

4.5. False-Positive Analysis and Threshold Sensitivity

On 100 unwatermarked cloaked faces decoded against random messages (500 trials), the BER of false detections concentrates around 0.50 (mean 0.496–0.500). Under the matched deployment condition (JPEG75 on both positives and negatives), Fig. 6 reports the TPR/FPR trade-off as a function of the detection threshold θ : at $\theta = 0.25$, TPR is 99.2% (low B), 98.6% (low C), and 100% (mid) at FPR 0.2–0.6%; tightening to $\theta \leq 0.15$ yields 0% FPR in 500 trials while retaining $\text{TPR} \geq 93\%$ (Table 4). This makes the detector highly specific and lets practitioners trade TPR against FPR by choosing θ .

Table 4: BER threshold sensitivity under deployment JPEG75: TPR (watermarked cloaked faces) vs. FPR (unwatermarked, 500 trials).

θ	TPR B (low)	TPR C (low)	TPR B (mid)	FPR (low / mid)
0.15	95.2%	93.1%	95.0%	0.00% / 0.00%
0.20	98.7%	97.4%	100%	0.20% / 0.00%
0.25	99.2%	98.6%	100%	0.20% / 0.60%
0.30	99.4%	98.9%	100%	0.40% / 1.80%

Table 5: Strong cloaking (Fawkes mid, n=200). WM = detection TPR under $\text{BER} \leq 0.25$.

Group	PSR $\uparrow$	WM clean	WM JPEG80	WM JPEG50	sim $\downarrow$
A: cloak (no WM)	95.0%	–	–	–	0.054
B: +WAM global	94.0%	100%	100%	93.5%	0.114
C: +WAM safety-region	93.5%	100%	100%	95.0%	0.083

4.6. Stronger Cloaking (Fawkes mid)

Under stronger cloaking (75-iteration, dual-extractor Fawkes), Table 5 shows that watermarking coexists losslessly with the cloak on 200 LFW faces: PSR remains 93.5–95% (vs. 95% for cloaking alone), and watermark detection reaches 99.5–100% (clean/JPEG80) and 93.5–95% (JPEG50) under $\text{BER} \leq 0.25$. Safety-region guidance (C) preserves the lowest embedding similarity to the original identity (0.083 vs. 0.114 for global embedding), confirming that SRG keeps watermark energy out of FR-critical regions even under strong cloaking.

4.7. Comparison with a SOTA Neural Watermarking Baseline

We compare against InvisMark [10] (WACV 2025), a representative SOTA neural watermarking method with a released checkpoint, using its official pre-trained model (100-bit messages, 256×256) applied to the same cloaked faces under the identical JPEG protocol (Table 6). InvisMark preserves cloaking well (PSR 94.5%) and decodes on clean images (bit accuracy 95.1%, TPR 100% under $\text{BER} \leq 0.25$), but its published checkpoint fails under JPEG compression even on *clean* faces (bit accuracy drops to $\approx 50\%$, i.e. random, at JPEG80/50/95), while our method maintains 93–100% TPR under JPEG80/50. This confirms that JPEG robustness is not an incidental property of neural watermarking and motivates the BER-based detection of

Table 6: Comparison with InvisMark (WACV 2025) under mid cloaking (n=200). WM = detection TPR under $\text{BER} \leq 0.25$; bit acc. = mean bit accuracy.

Method	PSR $\uparrow$	WM clean	WM JPEG80	WM JPEG50	PSNR/SSIM
B: WAM global ($\beta=6$)	94.0%	100%	100%	93.5%	29.0/0.851
C: WAM safety-region ($\beta=6$)	93.5%	100%	100%	95.0%	30.1/0.879
InvisMark (official ckpt)	94.5%	100% (95.1%)	0.0% (49.7%)	0.0% (50.2%)	49.6/0.991

Table 7: Sequential (C) vs. joint (E) under mid cloaking (n=20).

Pipeline	PSR	WM detection ($\text{BER} \leq 0.25$)
C: sequential (SRG-guided WAM)	85%	100%
E: joint optimization	85%	100%

our framework. Stable Signature [9] targets generative-model rooting and requires a latent diffusion decoder plus per-key extractor training, which is not directly applicable to arbitrary cloaked photographs; we therefore compare against InvisMark, the closest applicable baseline.

4.8. Sequential vs. Joint Optimization

We implement watermark-aware joint refinement (Sec. 3): starting from the cloaked image, we optimize the image tensor to minimize a weighted sum of an FR-retention loss (keeping the embedding distant from the original identity) and a WAM decoding loss (making the extractor read the target message). Table 7 shows that our sequential pipeline (C) is competitive with joint optimization (E) on both PSR and watermark detection, directly informing the sequential-vs-joint debate of EIAW [14] and ARFP [15]: sequential embedding is viable when combined with SRG and strength calibration.

4.9. Ablation: SRG Blending Strength α

We sweep the SRG blending factor α in Eq. 2 under mid cloaking (n=200, $\beta=6$; Table 8). Below $\alpha = 0.5$ the watermark is too weak to decode under JPEG50 (TPR 20.5% at $\alpha=0.25$); above $\alpha = 1.0$ cloaking retention degrades (PSR drops from 93.5% to 92.0–92.5%). The plateau at $\alpha \in [0.75, 1.0]$ (PSR 93.5%, WM 91–95% at JPEG50) confirms that $\alpha=1.0$ is near-optimal and that SRG is insensitive to small strength variations.

Table 8: Ablation of SRG blending strength α under mid cloaking ($n=200$, $\beta=6$).

α	PSR $\uparrow$	WM clean	WM JPEG50	PSNR (dB)
0.25	92.5%	35.0%	20.5%	41.9
0.50	93.0%	98.0%	75.0%	36.1
0.75	93.5%	100%	91.0%	32.5
1.00	93.5%	99.5%	95.0%	30.1
1.25	92.0%	100%	96.0%	28.2
1.50	92.5%	100%	97.5%	26.6

Table 9: Embedding order under mid cloaking. WM TPR under $\text{BER} \leq 0.25$ on clean (no JPEG).

Order	n	PSR $\uparrow$	WM clean	mean bit acc.
Cloak only (A)	200	95.0%	–	–
Cloak $\rightarrow$ WAM (C)	200	93.5%	99.5%	94.7%
WAM $\rightarrow$ cloak ($\beta=2$)	50	70.0%	100%	88.3%
WAM $\rightarrow$ cloak ($\beta=6$)	200	81.5%	100%	95.6%

4.10. Embedding Order: Watermark-Then-Cloak vs. Cloak-Then-Watermark

A natural alternative is to embed the watermark on the clean face and then cloak (watermark-then-cloak). Table 9 shows that this ordering is *inferior* for cloaking: the watermark perturbation consumes part of the perceptual budget, weakening the subsequent cloak. Under mid cloaking, watermark-then-cloak reduces PSR from 95.0% (cloak only) to 70.0% ($\beta=2$) and 81.5% ($\beta=6$), whereas our cloak-then-watermark pipeline preserves PSR at 93.5%. The watermark itself remains decodable in both orderings, confirming that the sequential direction is a design choice that matters for privacy preservation.

4.11. Multi-Message Capacity

With 32-bit random messages, decoding a watermarked image with its *own* message achieves 99.6% mean bit accuracy, while decoding with messages from other images yields 0% false positives in 87 cross-image trials—i.e., distinct messages do not interfere and the scheme supports independent per-image attribution.

4.12. Transferability and Limitations

Fawkes cloaking transfers poorly to FR models in distant feature spaces: under our protocol, cloaked faces remain correctly identified by arcface34 (PSR $\approx 0\%$) and FaceNet (PSR $\approx 0\%$), while they are protected against the training extractor (PSR 49–85% depending on mode). This is consistent with the model-specific nature of adversarial cloaking and is a limitation of the cloaking method rather than of our watermarking framework; we report cloaking retention under the training extractor (following the convention of Fawkes [1]) and discuss transferability as future work. Exact decoding (TPR@95%) under JPEG50 remains $\sim 10\%$ even at $\beta=6$; BER-based detection (Sec. 3) is therefore required, matching standard watermarking practice.

5. Conclusion

We studied provenance-verifiable privacy cloaking for face images. Our experiments establish four findings: (i) the deployment bottleneck of watermarking cloaked faces is JPEG recompression, to which cloaked faces are measurably more fragile (TPR@95% 99.4% $\rightarrow$ 6.0% under JPEG75; 85.2% vs. 88.3% bit accuracy vs. clean), while cloaking alone is harmless at the pixel level; (ii) safety-region guidance and strength-calibrated embedding with BER-based detection jointly restore watermark detectability to 93–100% at 0–0.6% FPR while preserving cloaking retention; (iii) under strong cloaking, sequential embedding is competitive with joint optimization and outperforms the reverse embedding order; and (iv) a SOTA neural watermarking baseline (InvisMark) fails under JPEG on the same protocol, underscoring the role of BER-based detection. These results provide the first practical evidence that adversarial face cloaking can be made provenance-verifiable without sacrificing privacy protection, and they inform the sequential-vs-joint debate in adversarial watermarking.

Acknowledgments

TBD.

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